

Цель работы

Целью данной лабораторной работы является ознакомление с работа с структура документа LaTeX.

Задание

1. Try adding text to your first document, typesetting and seeing the changes in your PDF.
2. Make some different paragraphs and add variable spaces.
3. Explore how your editor works; click on your source and find how to go to the same line in your PDF.
4. Try adding some hard spaces and see how they influence line-breaking.

Теоретическое введение

```
\documentclass{article}
\usepackage[T1]{fontenc}
\begin{document}
Hey world!
This is a first document.
\end{document}
```

(см. Рис. [-@fig:001]).

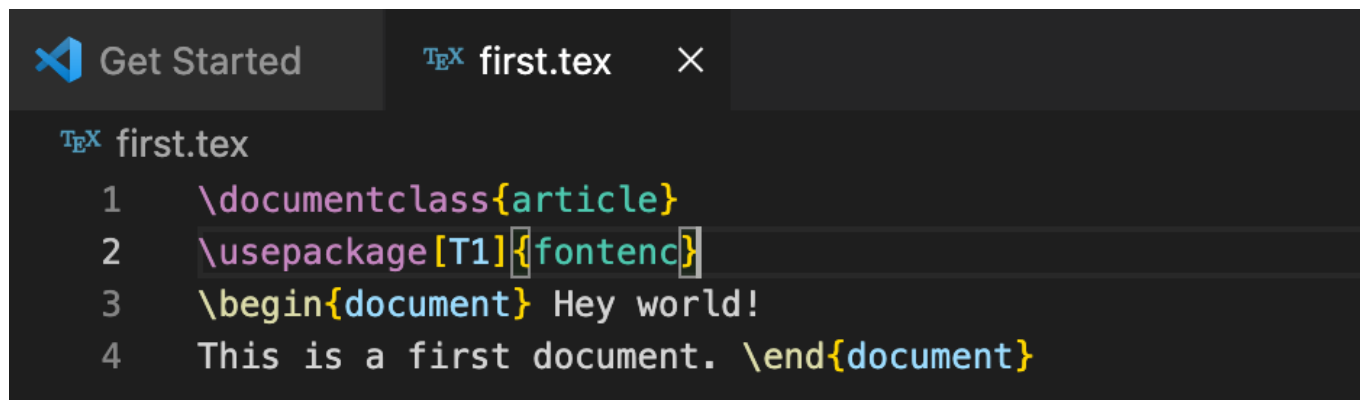
```
[nadiaezzakate@Nadias-MacBook-Air latex-lab % pdflatex first.tex
This is pdfTeX, Version 3.141592653-2.6-1.40.27 (TeX Live 2025) (preloaded format=pdflatex)
 restricted \write18 enabled.
entering extended mode
(./first.tex
LaTeX2e <2024-11-01> patch level 2
L3 programming layer <2025-01-18>
(/usr/local/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2024/06/29 v1.4n Standard LaTeX document class
(/usr/local/texlive/2025/texmf-dist/tex/latex/base/size10.clo)
(/usr/local/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(/usr/local/texlive/2025/texmf-dist/tex/latex/l3backend/l3backend-pdfTeX.def)
No file first.aux.

[1{/usr/local/texlive/2025/texmf-var/fonts/map/pdftex/updmap/pdftex.map}{/usr/l
ocal/texlive/2025/texmf-dist/fonts/enc/dvips/cm-super/cm-super-t1.enc}]
(./first.aux) </usr/local/texlive/2025/texmf-dist/fonts/type1/public/cm-super/
sfrm1000.pfb>
Output written on first.pdf (1 page, 15651 bytes).
```

{ #fig:001 width=100% }

Running LaTeX

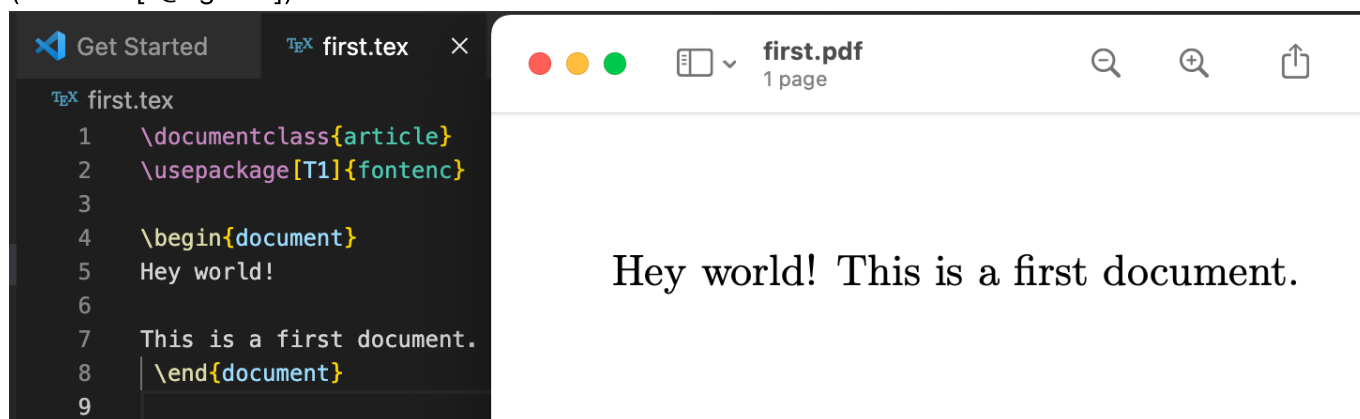
(см. Рис. [-@fig:002]).



```
Get Started  first.tex ×  
first.tex  
1 \documentclass{article}  
2 \usepackage[T1]{fontenc}  
3 \begin{document} Hey world!  
4 This is a first document. \end{document}
```

{ #fig:002 width=100% }

(см. Рис. [-@fig:003]).



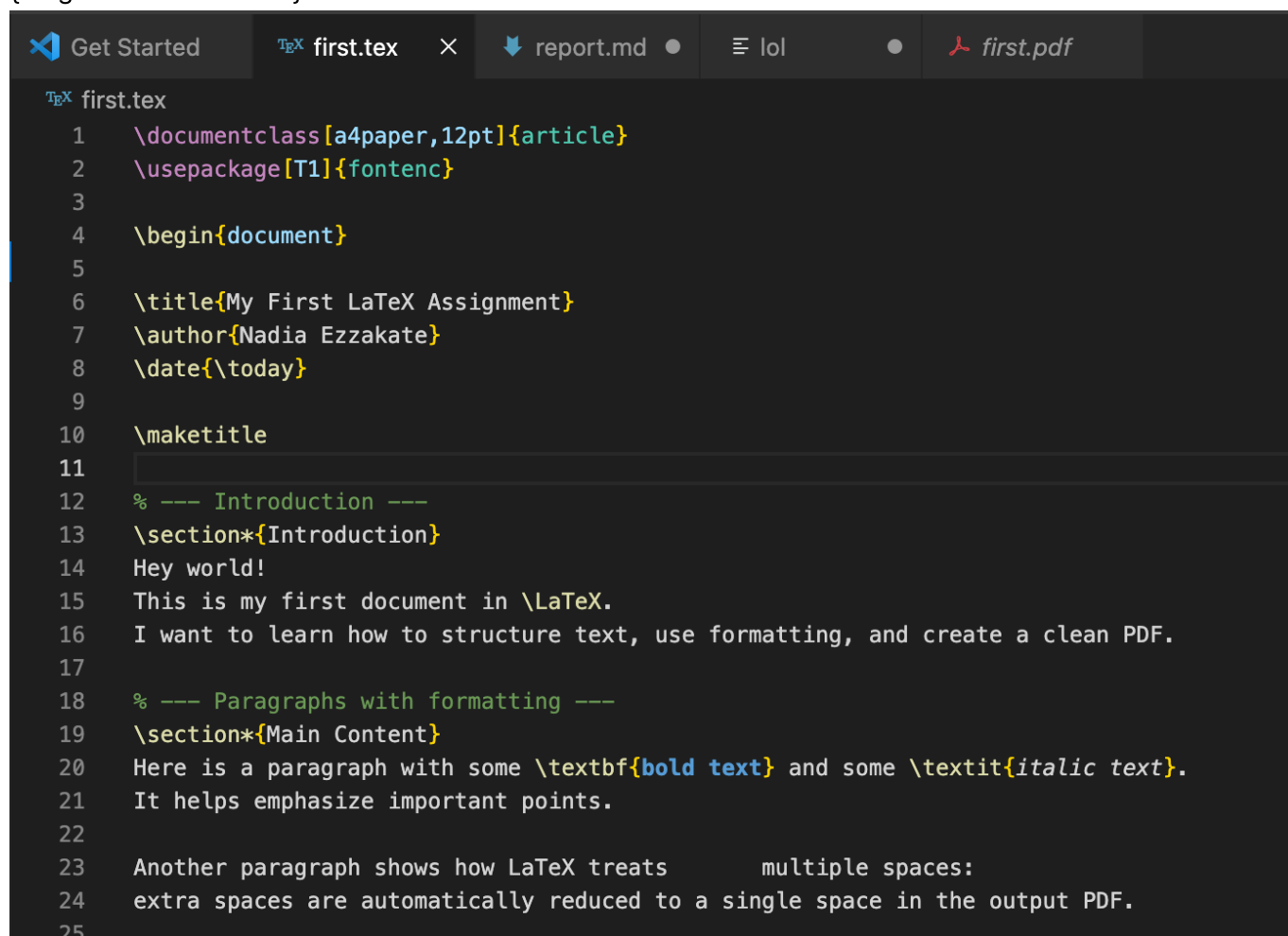
{ #fig:003 width=100% }

Выполнение лабораторной работы

Try adding text to your first document, typesetting and seeing the changes in your PDF.

(см. Рис. [-@fig:004]).

{ #fig:004 width=100% }



```

Get Started  TeX first.tex  ×  report.md  lol  first.pdf

TeX first.tex
1  \documentclass[a4paper,12pt]{article}
2  \usepackage[T1]{fontenc}
3
4  \begin{document}
5
6  \title{My First LaTeX Assignment}
7  \author{Nadia Ezzakate}
8  \date{\today}
9
10 \maketitle
11
12 % --- Introduction ---
13 \section*{Introduction}
14 Hey world!
15 This is my first document in \LaTeX.
16 I want to learn how to structure text, use formatting, and create a clean PDF.
17
18 % --- Paragraphs with formatting ---
19 \section*{Main Content}
20 Here is a paragraph with some \textbf{bold text} and some \textit{italic text}.
21 It helps emphasize important points.
22
23 Another paragraph shows how LaTeX treats      multiple spaces:
24 extra spaces are automatically reduced to a single space in the output PDF.
25

```

{ #fig:004 width=100% }

```

\documentclass[a4paper,12pt]{article}
\usepackage[T1]{fontenc}

\begin{document}

\title{My First LaTeX Assignment}
\author{Nadia Ezzakate}
\date{\today}

\maketitle

% --- Introduction ---
\section*{Introduction}
Hey world!
This is my first document in \LaTeX.
I want to learn how to structure text, use formatting, and create a clean
PDF.

% --- Paragraphs with formatting ---
\section*{Main Content}
Here is a paragraph with some \textbf{bold text} and some \textit{italic
text}.
It helps emphasize important points.

```

Another paragraph shows how LaTeX treats multiple spaces: extra spaces are automatically reduced to a single space in the output PDF.

% --- Hard spaces ---

Sometimes we need words to stay together:

Dr.~Brown lives in New~York~City.

The tilde (~) keeps these words on the same line.

% --- Lists ---

Here is a small list of things I want to remember about LaTeX:

\begin{itemize}

 \item Paragraphs are separated by blank lines.

 \item Special characters like \%, \\$, \#, _ must be escaped.

 \item Footnotes add extra information without breaking the text\footnote{This is an example footnote.}.

 \item Hard spaces (~) prevent unwanted line breaks.

\end{itemize}

% --- Conclusion ---

\section*{Conclusion}

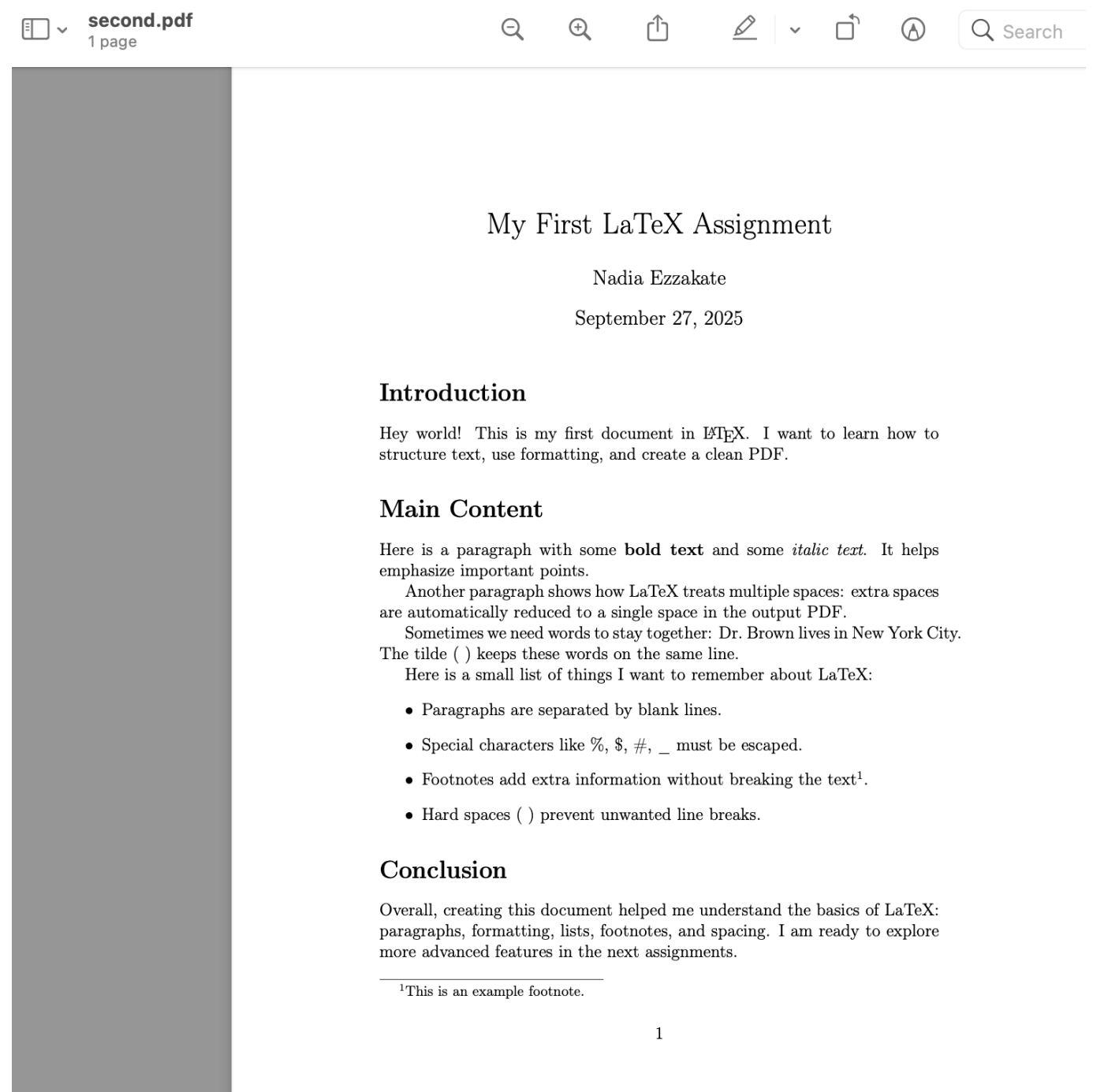
Overall, creating this document helped me understand the basics of LaTeX: paragraphs, formatting, lists, footnotes, and spacing.

I am ready to explore more advanced features in the next assignments.

\end{document}

Make some different paragraphs and add variable spaces.

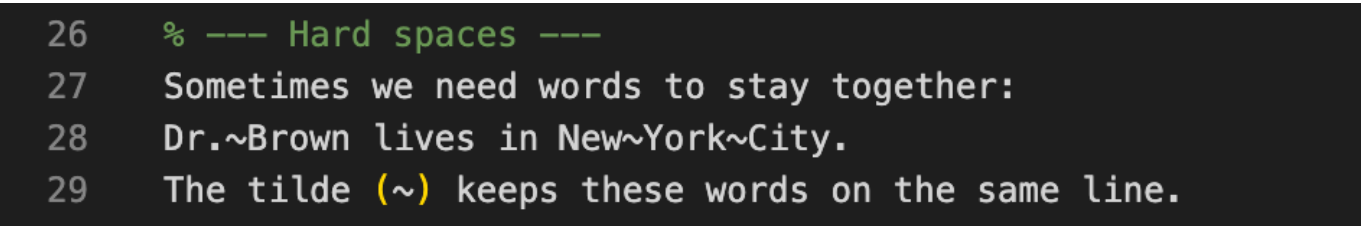
(см. Рис. [-@fig:005]).



{ #fig:005 width=100% }

Explore how your editor works; click on your source and find how to go to the same line in your PDF.

Try adding some hard spaces and see how they influence line-breaking.



{ #fig:005 width=100% }

(см. Рис. [-@fig:006]).

Sometimes we need words to stay together: Dr. Brown lives in New York City.
The tilde (~) keeps these words on the same line.

{ #fig:005 width=100% }